

WDSL WebToolkit

3-Minute Presentation Script

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Final Year Project 2026

SLIDE 1

~ 20 seconds

Title Slide

Good [morning/afternoon]. I'm Naveen Jayawardana, and my final year project is WDSL WebToolkit — an AI-powered platform that analyses websites for accessibility issues and presents the results to both developers and their non-technical clients through a shared workspace.

■ One breath. Move straight to slide 2.

SLIDE 2

~ 35 seconds

Introduction

Over 1.3 billion people worldwide live with a disability, yet more than 96% of the top websites have detectable accessibility failures. Tools like Lighthouse and axe DevTools exist — but they produce technical output that only developers can interpret.

Clients are left out entirely. And that communication gap between developers and their clients is something no existing tool addresses. That's the problem this project solves.

■ Keep this punchy — two facts, one gap, move on.

SLIDE 3

~ 40 seconds

Problem Statement & Objectives

The core problem is that developers and clients lack a single platform that gives both a detailed technical report and a clear, business-friendly explanation — from the same analysis.

My objective is to build exactly that. The four specific goals are: detect WCAG violations using axe-core and Lighthouse; convert those violations into plain English for clients using AI; give developers a technical report with highlighted screenshots; and connect both parties through a real-time Collaboration Hub.

■ Say the four objectives at a steady pace — don't rush them.

Proposed Solution & Methodology

The system works in four steps. A user submits a URL. axe-core and Lighthouse scan it and return structured violation data. An AI suggestion generator converts that data into plain English — so instead of "element missing accessible name," a client reads "some buttons on your site can't be understood by screen readers." Developers see the full technical report with screenshots. And both parties collaborate inside the built-in workspace.

The stack is React and Tailwind on the frontend, Python and Flask on the backend, with Playwright handling automated browser interactions.

■ The "instead of X, a client reads Y" line is your strongest moment — deliver it clearly.

Expected Outcomes & Timeline

The end result is a fully functional web app that makes professional accessibility analysis affordable for freelancers and small agencies worldwide — and helps build a more inclusive web.

Development runs from May through August 2026, ending with testing and documentation. Thank you — I'm happy to take questions.

■ Slow down on the last sentence. End clean — no "yeah, so, that's about it."

Total estimated time: ~ 2 min 50 sec – 3 min

Slide 4 is your core — practise it the most.